

Vidush Singhal

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West Lafayette, IN, USA

Summary

I am a Ph.D. student at Purdue ECE specializing in compiler techniques for program transformations using static analysis and whole-program analysis. My interests include compilers, data layout optimization, SIMD vectorization, automatic parallelization, compilers for GPUs, compilers for AI/ML, compilers for QC, programming languages, and high-performance computing. I have experience building compiler infrastructure in LLVM, Clang, MLIR, and Gibbon.

Education

Purdue University, West Lafayette, IN, USA

Direct Ph.D. in Electrical & Computer Engineering

Expected May 2027

Advisor: Prof. Milind Kulkarni

M.S. in Electrical & Computer Engineering (en route to Ph.D.)

Non-thesis, course-only

B.S. in Computer Engineering

May 2021

Selected Coursework: Compiler Code Generation and Optimization, Compilers for GPUs, Programming Parallel Machines, Reasoning About Programs, Programming Languages, Object-Oriented Programming in C++, Graph Theory, Applied Algorithms, Software Security, Intro To Quantum Computing

Experience

Purdue University - PLCL Group

West Lafayette, IN

Graduate Research Assistant; Advisor: Prof. Milind Kulkarni

May 2021 – Present

• Gibbon Compiler

- Designed and implemented compiler passes for optimizing the layout of recursive datatypes.
- Enhanced runtime performance of programs by improving the access patterns and spatial locality.
- Measured **speedups of 1.14x to 54x** over the best prior work.
- Implemented an array-of-structs to structure-of-arrays-style transformation for packed algebraic datatypes.
- Enabled selective traversal with **1.1x to 12x speedups** and optimizations such as loopification and vectorization.
- **Researched** on Gibbon's potential to accelerate ML inference and its execution on ML hardware such as GPUs.

• Orchard Compiler

- Developed a compiler framework to automatically fuse and parallelize tree traversals.
- Measured **1x to 5x speedups** over baseline tree traversals.

• Cornucopia

- Developed a fuzzing-based compiler framework to generate a corpus of diverse binaries from source programs.
- Used llc compiler optimization flags to drive differences in binaries.
- Discovered bugs in LLVM compiler optimization passes.
- Uncovered decompilation failures in Angr, Ghidra, and IDA binary analysis tools.
- Embedded ground-truth metadata in generated binaries to evaluate binary analysis tools.

• Copse Compiler

- Developed a compiler for vectorized evaluation of fully homomorphic encryption (FHE) decision trees.

Microsoft Research (RiSE)

Redmond, WA

Compilers Research Intern; Mentor: Dr. Angélica Moreira

May 2025 – Aug 2025

- Built C2Pulse, an LLVM/Clang-based transpiler for verifying memory correctness of annotated C programs.
- Implemented a Clang-to-Clang source transformer (~7K LOC of C++) that emits Pulse code for F* verification.
- Built a C benchmark suite and verified C components of DPE (Dice Protection Environment) through C2Pulse.

Lawrence Livermore National Laboratory (LLNL)

Livermore, CA

Computing Scholar Intern; Mentor: Dr. Johannes Doerfert

May 2024 – Aug 2024

- Engineered an LLVM Attributor pass to optimize byte layout within stack allocations for improved spatial locality.

- Implemented LLVM IR analyses and transformations to eliminate dead bytes and reorder frequently accessed bytes.
- Optimized the LLVM GPU sanitizer and reduced XSBench runtime by **6%** by hoisting checks out of for-loops.

Nod.ai Labs (Acquired by AMD)

Machine Learning Compiler Engineering Intern; Supervisor: Anush Elangovan

Santa Clara, CA
May 2022 – Aug 2022

- Implemented Torch-MLIR kernels and compiler support for multiple PyTorch operators.
- Improved the PyTorch-to-MLIR compilation flow by expanding operator coverage.
- Added support for the Facebook DLRM recommendation model in the PyTorch-to-MLIR pipeline.
- Investigated ILP solver improvements for the Alpa scheduling framework.

Senior Design Project - SoCET Team

B.S. Final Project; advised by Dr. Mark Johnson and Prof. Samuel Midkiff

West Lafayette, IN
Jul 2020 – May 2021

- Developed an LLVM-based compiler tool for proprietary RISC-V hardware.
- Implemented multiple LLVM RISC-V backend Machine IR passes to identify sparse program regions.
- Annotated sparse regions so hardware could detect them at runtime and perform runtime checks.

Selected Projects

- **Global Tensor Layout Optimization for IREE**
Researched tensor layout optimization in the `iree-global-optimization-opt-level` compiler pipeline.
- **LLM-Assisted Specification Inference for Verus**
[Explored](#) LLM-assisted inference of Verus specifications using symbolic reasoning tools like Seahorn and Crucible.
- **Explored OpenMP GPU Offloading for Parallel STL-style Algorithms**
[Tested](#) a proof-of-concept evaluation of OpenMP GPU offloading for handwritten algorithms to validate speedups.

Publications

- **Vidush Singhal**, Mikah Kainen, Artem Pelenitsyn, Michael H. Borkowski, Mike Vollmer, Milind Kulkarni. “SoCal: A Language for Memory-Layout Factorization of Recursive Datatypes.” [Preprint under submission, 2026](#).
- Raghav Malik, **Vidush Singhal**, Benjamin Gottfried, Milind Kulkarni. “Vectorized Secure Evaluation of Decision Forests.” [PLDI 2021](#).
- **Vidush Singhal**, Akul Abhilash Pillai, Charitha Saumya, Milind Kulkarni, Aravind Machiry. “Cornucopia: A Framework for Feedback Guided Generation of Binaries.” [ASE 2022](#).
- **Vidush Singhal**, Chaitanya Koparkar, Joseph Zullo, Artem Pelenitsyn, Michael Vollmer, Mike Rainey, Milind Kulkarni. “Optimizing Layout of Recursive Datatypes with Marmoset.” [ECOOP 2024](#).
- **Vidush Singhal**, Laith Sakka, Kirshanthan Sundararajah, Ryan Newton, Milind Kulkarni. “Orchard: Heterogeneous Parallelism and Fine-grained Fusion for Complex Tree Traversals.” [ACM TACO 2024](#).
- Chaitanya S. Koparkar, **Vidush Singhal**, Aditya Gupta, Mike Rainey, Michael Vollmer, Artem Pelenitsyn, Sam Tobin-Hochstadt, Milind Kulkarni, Ryan R. Newton. “Garbage Collection for Mostly Serialized Heaps.” [ISMM 2024](#).

Skills

Languages:

Experienced: C, C++, Haskell, Python

Familiar: Java, Bash, CUDA, LLVM-IR, MLIR, MATLAB, Assembly (ARM, x86), SystemVerilog

Compilers & Verification: LLVM, Clang, MLIR/Torch-MLIR, IREE, GCC, Coq, Dafny, F*, Pulse

Parallel, Systems & ML: OpenMP, MPI, Pthreads, OpenCilk, Intel TBB, gem5, AFL++, GDB, Linux, PyTorch, TensorFlow, NumPy

Awards, Grants & Service

Dean’s List and Semester Honors, Purdue University (2017 - 2020)

LLVM Developers’ Meeting Travel Grant 2024

Research Service - AEC/PC/Reviewer for PLDI, CGO, ICFP, PPOPP, POPL, SPLASH, C3PO, JOSS (2023–2026)

Seminar Co-Coordinator, [Purdue Programming Languages and Systems Research Group](#) (2025 - present)